talenciaglobal

Number of Islands

Graphs, Matrix as Graph, Flood Fill · Mentor-Led Lab (Student Edition)

Developed by TalenciaGlobal

Difficulty ★★★ Advanced

Problem

Treat a grid as a graph: each 1 is land connected to its four orthogonal neighbours. Count the islands (4-connected groups of land) and report the size of the largest.

Implement the flood fill (BFS or DFS) over the grid from scratch.

What to Compute

- Islands: the number of 4-connected groups of 1s.
- Largest: the number of cells in the largest island.

Input / Output Format

Input: Line 1: R C. Next R lines: C integers each (1 = land, 0 = water).

Output: Two lines: Islands and Largest.

Examples

Example 1

Input: 4 5 1 1 0 0 0 1 1 0 0 1 0 0 0 1 1 0 0 0 0 0

Output:

```
Islands: 2
Largest: 4
```

Explanation: A 2x2 block of land (size 4) in the top-left and an L-shaped island (size 3) on the right give two islands; the largest has 4 cells.

Example 2

Input: 2 2 1 0 0 1

Output:

```
Islands: 2
Largest: 1
```

Explanation: Two diagonal land cells are not 4-connected, so they form two separate islands of size 1.

Constraints

- Cells connect up, down, left and right (not diagonally).
- Scan cells in row-major order.
- An island is a maximal 4-connected group of 1s.

Sample Run

Sample Input

```
4 5
1 1 0 0 0
1 1 0 0 1
0 0 0 1 1
0 0 0 0 0
```

Sample Output

```
Islands: 2
Largest: 4
```